

- The reaction catalysed by catalase is shown in Fig. 6.1.



- In the first experiment, hydrogen peroxide solution was added to three of the pieces of potato tissue and the progress of the reaction was measured.

The diagram illustrates an experimental setup for measuring the rate of reaction. A syringe containing a H_2O_2 solution is connected to a flask containing a potato. The flask is placed in a water bath. A delivery tube leads from the flask to a graduated cylinder, which is inverted in the water bath to collect the gas produced. The setup is used to measure the volume of gas produced over time.

Fig. 6.2

- [2]

- (ii) In the second experiment, the student cut each of the three remaining pieces of potato to obtain six pieces, each measuring 10 mm $\square$ 10 mm $\square$ 10 mm.

Using exactly the same conditions, the student measured the progress of the reaction and obtained different results to the first experiment.

Explain why the results of the second experiment were different from the results of the first experiment.

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..... [2]

- (b) The student then investigated the effect of temperature on the activity of catalase.

On Fig. 6.3, sketch a curve to show how temperature affects the activity of an enzyme such as catalase.

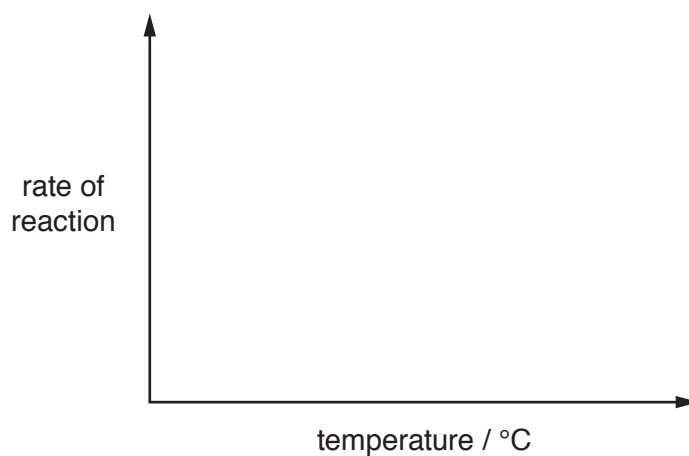


Fig. 6.3

[1]

[Total: 5]